

Michael Milowski
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U.S. Citizen

I am an aerospace engineering student with strong leadership, team collaboration, and communication skills. I work effectively in teams presenting complex information clearly while supporting team members to achieve goals. I am detail oriented, quick to learn new tools, and skilled at applying technical knowledge to practical problems. My long-term professional goal is to apply my knowledge to support space missions and help drive the future of space exploration.

Education

University of Illinois Urbana-Champaign , Champaign, Illinois	August 2024 – May 2028
Bachelor of Science in Aerospace Engineering	
GPA: 3.84/4.00	Credit hours completed: 101
Lake Zurich High School , Lake Zurich, Illinois	August 2020 – May 2024
Graduated Magna Cum Laude	
GPA: 4.50/4.00	
High Honor Roll (2020-2024)	Illinois State Scholar 2024
AP Scholar with Distinction – Received the highest AP award for excellence across eight exams	
Extracurricular Activities: Best Buddies, Interact, Habitat for Humanity, Finance Club, Basketball, Tennis	

Experience

Deputy Project Manager of Resources NASA L'SPACE Program	January 2025 – May 2025
Hours Worked Each Week: 15	

The NASA L'SPACE program tasked a multidisciplinary team of students with creating a comprehensive NASA mission concept. I was elected to manage the team's programmatic strategy, a responsibility that required me to ensure accurate funding allocation, establish realistic phase timelines, and conduct comprehensive risk management. This leadership role allowed me to guide the team through the complex process of turning an abstract mission idea into a structured, feasible design.

To guide the team, I led weekly meetings to coordinate goals, communicate progress, and plan key programmatic elements of the mission, while fostering open communication and problem-solving within a diverse multidisciplinary team. I strategically delegated tasks based on members' skills and interests, provided additional resources to support their work, and maintained an open line of communication that enhanced both individual performance and overall team effectiveness. Effective communication not only improved our mission concept but allowed the team to develop meaningful connections.

I supported the team's work by learning and applying a variety of professional tools used in NASA projects, including Gantt charts, risk matrices, the NASA Instrument Cost Model, and the Mission Operations Cost Estimation Tool. Learning to use these tools and guiding others in their use allowed the team to better understand the NASA mission life cycle and federal standards. Maintaining organized documentation that was available to everyone on the team ensured transparency, improving accuracy and resulting in a cohesive final report.

Under my leadership as Deputy Project Manager of Resources, the team produced a holistic preliminary design review that demonstrated feasibility, cost effectiveness, and risk analysis at a level consistent with NASA expectations. The final report and presentation reflected organization, technical writing knowledge, and use of quantitative and qualitative analysis. I also attended and engaged in weekly lectures taught by NASA scientists, engineers, and managers using their guidance to influence my leadership style and technical approach. This experience enhanced my ability to analyze, design, and manage complex projects requiring collaboration, structured implementation, and rigorous evaluation.

NASA Human Lander Challenge (HuLC) Illinois Space Society

August 2024 – Present

Hours Worked Each Week: 10

As a member of the Illinois Space Society competing in NASA's Human Lander Challenge (HuLC), I developed technical expertise in cryogenic propellant transfer systems to address the problem of cryogenic propellant transfer in microgravity conditions. Part of our team's solution required that we minimize propellant loss; I focused on modifications for propellant transfer tubes that decrease line chill down time and reduce propellant boiloff.

To take on this challenge, I conducted comprehensive literature reviews of over 30 research papers pertaining to advanced cryogenic systems and applied my research to Artemis propellant transfer conditions. I presented this technical information in a clear and approachable manner, ensuring understanding throughout the team. I analyzed and evaluated multiple approaches creating a detailed trade matrix comparing tube modifications qualitatively and quantitatively. In collaboration with the team, I integrated my research into the development of a holistic proposal. In addition, I presented my research to an aerospace engineer at NASA's Glenn Research Center specializing in cryogenics and incorporated his feedback into the next phase of design to strengthen both feasibility and compliance with NASA standards.

By analyzing numerous methods that increase pipe chill down and mitigate boiloff, my proposed solution increases cooldown efficiency by up to 176% and produces a 65% savings of propellant during transfer. Our final concept, ECLIPSE (Efficient Cryogenic Low Invasive Propellant Supply Exchange), was selected as one of 12 national finalists in the HuLC competition. Our team received an award for the *Best Technical Presentation*, and I am currently working with a small team to publish a perspective paper based on our research.

Kappa Theta Tau - Professional Society for Engineering Students

2024 – Present

Hours Worked Each Week: 6

As a member of Kappa Theta Tau, I was selected from a field of over 400 students in recognition of my commitment to professionalism, philanthropy, diversity, and brotherhood. I actively participate in multiple events every week each designed to promote one or more of our values.

For example, I collaborated with a team of peers to plan, organize, and execute three successful fundraising events raising over \$1,000 for charity while developing effective communication and project management skills. I also orchestrated a student STEM event at the Champaign Public Library to promote engineering in a fun and informative way by developing and leading a lesson and activity for children in afterschool care. In addition, I work with a team to do hands-on engineering work creating projects for Illinois Engineering Open House. This is a year-long project in which a small team collaborated to ideate, design, build, and present a passion project aimed at both educating and entertaining students of all ages. I designed and manufactured multiple mechanical components for an interactive bus route display to showcase different physics principles.

My involvement with Kappa Theta Tau has allowed me to continue giving back to my community as well as develop professional skills. From helping potential new members with interview preparation, to meeting hundreds of new students and helping them navigate college, I have developed a deeper understanding of how to foster collaboration, inclusivity, and service in engineering.

Volunteer Lake Zurich High School

2021 – 2024

Hours Worked Each Week: 10

In high school, I wanted to become more involved in my community and joined multiple service organizations. Through the Best Buddies program, I built meaningful connections and friendships with students who have special needs through after-school activities and social events. These events allowed me to see the impact I made by being dependable, caring, and inclusive.

While in Interact Club, I volunteered over 60 hours supporting community services and organizations including food pantries, elementary and middle schools, and assisted living facilities, providing hands-on assistance to meet local needs. These volunteer opportunities allowed me to see firsthand how kindness and offering time and attention greatly impact people's lives.

In addition, I contributed to the construction and renovation of homes for underserved families, gaining hands-on experience in collaborative project work while promoting community development. Across these organizations, I participated in over 100 volunteer events, often taking on leadership roles to organize activities, guide peers, and ensure successful outcomes. These experiences helped build my work ethic and communication skills while creating a meaningful difference in my community.

Child Care Provider- Private Residence

2016 – 2021

Hours Worked Each Week: 10

I assisted in providing after-school care for multiple children under the age of five. In order to promote physical, emotional, and cognitive growth I created a safe and nurturing environment and planned and implemented age-appropriate activities. Additionally, I communicated the daily activities and other relevant information to parents. This experience strengthened my ability to manage responsibilities independently, enhanced my adaptability, and improved my skills in organization, communication, and problem-solving.

Projects

1:1 Model of NASA SLS Block 1

Completed: May 2025

I created a scale model of NASA SLS Block 1 using NX modeling software. I chose to create SLS because of its importance in the future of space exploration and large influence on my passion for aerospace technology. I used dozens of documents and reference pages along with photogrammetry to ensure accurate measurements. My model showcased my proficiency with NX, using complex assemblies as well as 3D solid and surface modeling. The model consisted of over 30 moving components including detailed representations of each detachable stage of SLS. To enhance the realism of my model, I used 48 unique materials and applied 43 decals to curved and complex surfaces. After completion, I used photorealistic rendering with advanced lighting techniques as well as an assembly motion animation with kinematic simulation. This project allowed me the opportunity to learn advanced techniques in NX as well as enhance my practical engineering skills such as interpreting documentation, research, and attention to detail.

Skills

CAD & Design Software: Siemens NX, AutoCAD, Autodesk Inventor, Fusion 360

Programming Languages: C++, Python, Java

Workspace Tools: Google Workspace, Microsoft Office Suite

Soft Skills: Leadership, Team Management, Collaboration, Communication, Presentation Skills, Research, Attention to Detail, Self-directed Learning, Technical Literacy, Project Management